

Deep learning in prostate cancer diagnosis and Gleason grading in histopathology images: An extensive study

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ABSTRACT

Among American men, prostate cancer is the cause of the second-highest death by any cancer. It is also the most common cancer in men worldwide, and the annual numbers are quite alarming. The most prognostic marker for prostate cancer is the Gleason grading system on histopathology images. Pathologists determine the Gleason grade on stained tissue specimens of Hematoxylin and Eosin (H&E) based on tumor structural growth patterns from whole slide images. Recent advances in Computer-Aided Detection (CAD) using deep learning have brought the immense scope of automatic detection and recognition at very high accuracy in prostate cancer like other medical diagnoses and prognoses. Automated deep learning systems have delivered promising results from histopathological images to accurate grading of prostate cancer. Many studies have shown that deep learning strategies can achieve better outcomes than simpler systems that make use of pathology samples. This article aims to provide an insight into the gradual evolution of deep learning in detecting prostate cancer and Gleason grading. This article also evaluates a comprehensive, synthesized overview of the current state and existing methodological approaches as well as unique insights in prostate cancer detection using deep learning. We have also described research findings, current limitations, and future avenues for research. We have tried to make this paper applicable to deep learning communities and hope it will encourage new collaborations to create dedicated applications and improvements for prostate cancer detection and Gleason grading.

1. Introduction

Prostate cancer is a serious disease worldwide. In fact, men of all ages can be affected by this deadly disease. Day to day, technology is spreading its branches to every sector, including the medical industry. Recently, the use of Computer-Aided Diagnosis (CAD) has been increased to help doctors make correct decisions. Early detection of rapid recognition plays a vital role in the diagnosis and prognosis of prostate cancer. Bio-medical imaging is very crucial for efficient cancer identification and treatment. It is quite challenging for pathologists to detect anomalies in biopsy reports quickly and efficiently. Manual processing takes a considerable amount of time and delays treatment. Moreover, manual processing is not cost-efficient and it is time-consuming. Deep learning can deliver improved Gleason grades while reducing human errors by increasing accuracy regardless of location. Deep learning techniques in medical imaging have already shown promising results.

From the early days, back from 1980, Computer-Aided Diagnosis

(CAD) has been used in different medical fields. In CAD applications that use medical imaging, machine learning methods are commonly used to detect cancer. In the last decade, machine learning and deep learning technology have improved significantly. Furthermore, this improvement also contributes to CAD applications. Deep learning can learn high-level features from the images. With the introduction of deep learning methods, it may be possible to achieve high detection accuracy without using hand-crafted features, as features may be extracted during training. Moreover, with the help of massively parallel computing (GPUs) in recent years, deep learning techniques have achieved immense popularity in prostate cancer detection and Gleason grading. The paper aims to present some conventional deep learning techniques and a complete overview of prostate cancer detection applications and Gleason grading. In short, our paper explains:

- A closer look into different histopathology image datasets and their sources

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- Image pre-processing, post-processing, and evaluation techniques and their limitations
- Analysis of existing significant deep learning methodological approaches and their impact in differentiating various prostate cancer Gleason grades
- Highlights of promising solutions that resolve obstacles faced in prostate cancer detection and Gleason grading
- Existing limitations, and future scope in this field

The remainder of this paper is as organized as follows. Section 2 introduces the basic concepts of prostate cancer. In section 3, we discuss different Gleason grading methods. Section 4 describes the different concepts of deep learning. It is primarily about Convolutional neural networks, which act as the backbone for image processing and analyzing image data for classification and segmentation. Section 5 discusses the image datasets available and frequently used by researchers around the world. Section 6 discusses the pre-processing and post-processing techniques essential for classification, segmentation, enhancement, and resolution reduction of images. Section 7 discusses the different challenges that researchers face while using prostate histopathological images. In section 8, we discuss different deep learning models used in recent times by different researchers. We summarize our whole study and future opportunities in section 9. In section 10, we draw a curtain to our paper.

2. Basic concepts of prostate cancer and its detection

Prostate cancer is the most common cancer in the United States and is the second most deadly in men [1]. To identify different kinds of prostate tumors, pathologists use different screening methods. Male hormones such as testosterone cause prostate cancer to grow and survive. Like all cancers, prostate cancer begins when a mass of cells has grown out of control and invades other tissues. Cells become cancerous due to the accumulation of defects, or mutations, in their DNA. Mutations in the abnormal cells' DNA cause the cells to grow and divide more rapidly than normal cells do. The abnormal cells continue living when other cells would die. Acinar adenocarcinoma of the prostate comprises 90–95% of prostate cancers diagnosed. Ductal carcinoma and neuroendocrine carcinoma account for the majority of additional cases.

The prostate is a small walnut-shaped gland in the male reproductive

system that surrounds the urethra below the bladder. It produces the seminal fluid that nourishes and transports sperm. As shown in Fig. 1, healthy prostate tissue consists of non-glandular stroma (fibromuscular tissue) and stroma-surrounding glands. These different tissues are tightly fused and surrounded by a joint capsule. Each gland unit consists of the lumen and epithelial cells. Carcinomas of the prostate arise most commonly in the outer, peripheral zone of the gland. Cells develop in and out of the gland in cancerous tissues, interrupting prostate glands' general structure and organization. Cancerous tissue has uncontrolled replication of epithelial cells that interrupt the regular arrangement of gland units. Epithelial cells usually substitute both stroma and lumen in high-grade cancer.

3. Gleason grading methods

The Gleason grading system is one of the most reliable methods for evaluating prostate cancer aggression [2], developed in 1967 and updated in 2014. Gleason grades are used for describing prostate adenocarcinoma growth patterns, and they are related to disease severity. According to this system, prostate cancers are scaled into five grades based on glandular patterns of differentiation. It varies from 1 (excellent prognosis) to 5 (poor prognosis). Deep learning technology can contribute significantly to the automatic detection of cancer in prostate tissues and predict the severity of the cancer stage.

3.1. Historical overview of Gleason scoring system

In the 1960s–1970s, a revolutionary research study initiated by the Veterans Administration of the United States started a Gleason system of grading prostate cancer. This study involved 2,900 patients for testing [59]. In that study report, Dr. Donald Gleason explained the histological growth patterns of prostate adenocarcinoma. Although several grading systems were available at that time, this study made a remarkable impact on the future of Gleason grading and its evolution.

In 2004, World Health Organization (WHO) approved the Gleason grading system. It has also been included in the AJCC/UICC staging system and NCCN guidelines. The classical Gleason grading system consists of five histological growth patterns, and the biopsies or whole slide images are graded accordingly. Gleason 1 describes the best differentiated, whereas Gleason 5 describes the least differentiated. Gleason 1 is more correlated and favorable for the prognosis. However, Gleason 5 is correlated with poor prognosis. According to the classical Gleason scoring system, if the tissue cells or glands are uniform and small, it is well-differentiated and classified into Gleason 1. Gleason 2 has more stroma and more spaces between the glands than Gleason 1 growth patterns. Gleason 3 has a distinct interpretation of cells from glands at margins. Gleason 4 has abnormal masses of cells with few glands. Gleason 5 has a lack of or irregular glands, and it looks like a sheet of cells.

The Gleason grading process consists of finding and classifying cancer tissue into Gleason patterns based on the tumor's architectural growth patterns. After marking a biopsy report into the corresponding Gleason score, it is converted into an ISUP grade on a 1–5 scale [58]. The Gleason grading method is unique because it relies entirely on tumor architectural features. The system focuses on cytological appearances and not on the worst pattern [75]. The Gleason scoring (GS) considers two common patterns. This system is one of the tremendous prognostic markers for prostate cancer detection. The ISUP grade system plays an integral part in the cancer detection of patients all around the world. Correctness in Gleason grading is vital to help pathologists in making the correct decision.

After 2014, homogenous Gleason grading has evolved. Heterogenous Gleason grading was introduced. In the heterogenous grading method, majority and minority growth patterns are taken from the biopsy. Adding the Gleason scores, a new Gleason grade is obtained. According to ISUP grade, a Gleason score of 3 + 3 is ISUP grade 1, 3 + 4 falls into

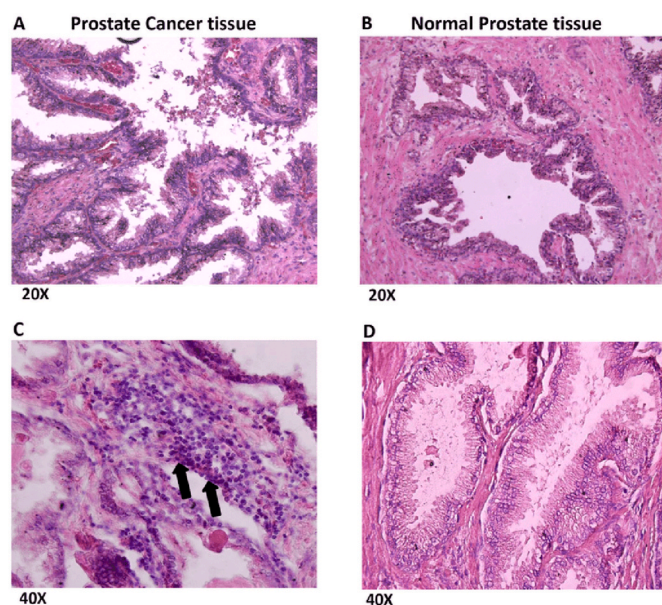


Fig. 1. Prostate Cancer tissue(left) vs Normal Prostate tissue(right) Arrows indicate infiltrating lymphocytes [115].

ISUP grade 2, 4 + 3 into 3rd, 4 + 4, 3 + 5, and 5 + 3 falls into 4th grade. Finally, 4 + 5, 5 + 4, and 5 + 5 fall into ISUP grade 5. Fig. 2 represents the Gleason scoring method.

3.2. Gleason scoring difficulties

The distinction between GP 3 patches and GP 4 patches is a difficult task. Researchers find GP 3 and GP 4 differentiating more problematic than other Gleason patterns [76,77]. Mainly fused or small Lumia-free glands may be classified into either GP 3 or GP 4 [77].

4. Overview and basic concepts of deep learning techniques

4.1. Core components of CNN

CNN is a type of neural network that is helpful for detection, classification, and other processing of images. Convolutional neural networks (ConvNets or CNNs) [5–8] are also deep feed-forward networks, but the only difference is that adjacent layers are connected in different ways. CNN architectures stack a few convolutional layers, pooling layers. For networks going deeper to deeper, the image becomes smaller to smaller, but more feature maps are provided. The purpose of different elements in CNN architecture has been discussed concisely in the following section.

4.1.1. Convolutional layer

The convolution layer is the basic building block of CNN. The convolution layer has many kernels. Each of the neurons acts as a kernel. Different kinds of kernels/filters can perform operations on images such as edge detection, blur, and sharpen by applying convolution. In the convolution kernel, images are split into small pieces and extract features from each small block. The kernel uses a specific set of weights to communicate with images by multiplying their elements by the receptive region's corresponding elements [10]. Furthermore, convolution can be classified into various forms depending on strides, filters, padding [11].

4.1.2. Pooling layer

The pooling layer aims to reduce the number of parameters when the image is too large and limit the overfitting risk. It also minimizes computational load, memory usage. Spatial pooling is often referred to as sub-sampling or down-sampling. The dimension of each map is reduced, but essential details are preserved. CNN uses different pooling formulations, including max pooling, sum pooling, average pooling, L2-norm pooling, overlapping, spatial pyramid pooling, etc. [12–14].

4.1.3. Activation function

At the end or in neural networks, an activation function or layer is a node which is a deciding function for learning intricate patterns. The choice of an effective activation function can accelerate the process of learning. In recent decades, Sigmoid and TanH functions have been used as the activation function. ReLU is currently the most used activation function in the world and is used in nearly all CNNs architectures. ReLU and modified versions help to solve the vanishing gradient problem [15, 16]. The ReLU's activation function is significantly computationally efficient as all neurons do not activate simultaneously. ReLU is converging six times faster in practice than TanH and Sigmoid.

4.1.4. Batch normalization

Batch normalization helps pace deep learning processes by raising the number of unit values by moving around hidden values (covariance shift). Also, batch normalization makes it much easier to learn more independently from each network layer. A layer's features are independently normalized with a mean zero and variance one in batch normalization [17,18].

4.1.5. Dropout

Dropout is a regularization technique proposed by Hinton [19] that ignores randomly chosen neurons during training. Dropout enables learning more complex features and approximately doubles the number of iterations required to converge.

4.1.6. Fully connected layer

The fully connected (FC) layer within CNN utilizes high-level

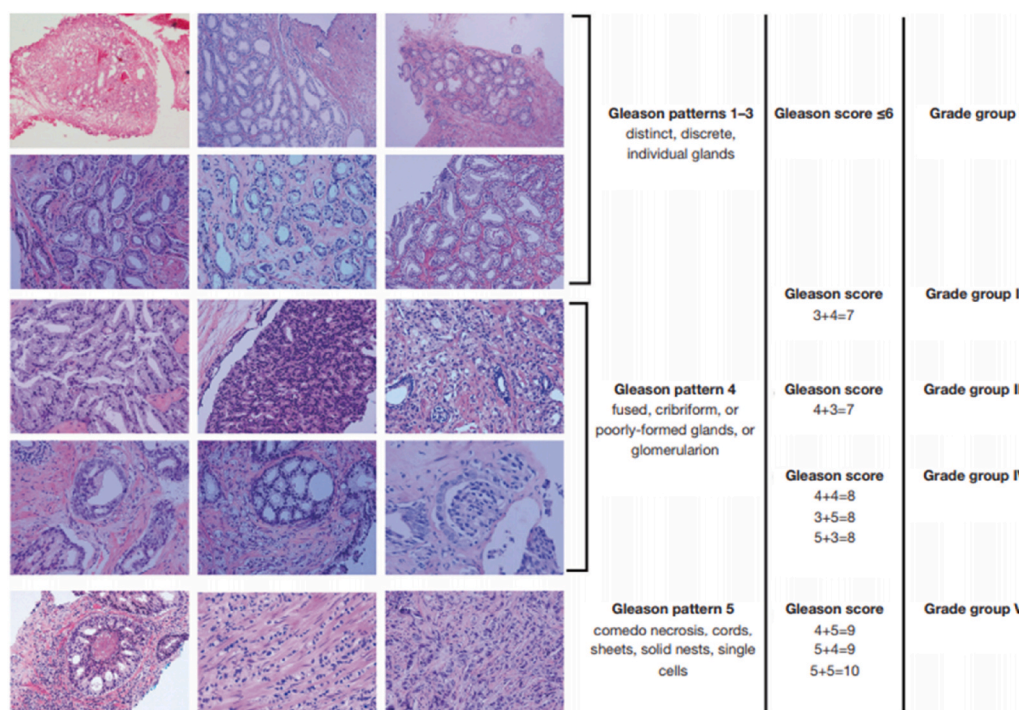


Fig. 2. Gleason scoring from the paper of Chen et al. [59].

features from convolution or pooling layers. The fully connected layer classifies the input image into various groups based on the dataset [21]. In a fully connected layer, softmax is mainly used as an activation function for classification, and the number of layers included in the network model is not controlled strictly.

4.1.7. Unpooling layer

During the pooling operation, create a matrix that records the maximum value's location, and the unpool operation will insert the pooled value in the original place, with the remaining elements being set to zero. Unpooling captures example-specific structures by tracing the original locations with strong activations back to image space. As a result, it effectively reconstructs the detailed structure.

4.2. Training method

4.2.1. Transfer learning

Transfer learning [22,23] is a machine learning approach where the pre-trained model reuses a new problem. It will not only considerably speed up training but also require substantially less training data. Transfer learning is a powerful tool when a neural network handles limited data for a new domain, and a sizeable pre-existing data pool can be transferred to the task. Labeled data sets are limited in medical imaging. Transfer learning is a perfect option for managing minimal medical data. Transfer learning strategies can be separated into two separate sections. They are:

Use the pre-trained model as a feature extractor: This technique uses a pre-trained model (like ImageNet) as a feature extractor to handle the convolutional neural network. This removes the last fully connected layer (classifier layer) and then treats the remaining layers for a new task. Instead of the entire network, this approach trains only a new classifier, which significantly speeds up training.

Fine-tuning: Another strategy is fine-tuning technique. The fine-tuning technique removes the final layer. It also retrains several previous layers selectively. The process is done by backpropagation. All CNN layers can be fine-tuned.

4.2.2. End-to-end learning

Most deep-learning systems have several implementation stages. However, the deep learning system integrates all these stages with a single neural network. This is a deep learning methodology that trains all parameters together. Parameters are being learned altogether in this process, not step-by-step. The only difference between the end-to-end learning process and the Deep Learning process is that the end-to-end learning process must collect all of the parameters jointly (at the same time), while the Deep Learning process can collect the parameters either jointly or step by step. Therefore, every end-to-end learning process is a Deep Learning process, but not every Deep Learning process is an end-to-end learning process.

4.2.3. Multi-task learning

Multi-task learning (MTL) is a subfield of machine learning in which multiple learning tasks are solved at the same time while exploiting commonalities and differences across tasks. This can improve learning efficiency and prediction accuracy for the task-specific models compared to training the models separately [87].

According to Rich Caruana et al. [87], MTL improves generalization by leveraging the domain-specific information contained in the training signals of related tasks. It does this by training tasks in parallel while using a shared representation. In effect, the training signals for the extra tasks serve as an inductive bias. The MTL net uses a shared hidden layer trained in parallel on all the tasks; what is learned for each task can help other tasks be learned better.

4.3. Innovation in CNN architecture

4.3.1. AlexNet

AlexNet [25] and LeNet [24] have very similar architecture. However, AlexNet is more in-depth, with more convolutional layers and more filters per layer. AlexNet has eight layers — 5 convolutional and three fully connected. For adding non-linearity, ReLU is being implemented after every convolutional and fully connected layer instead of TanH. AlexNet also uses dropout instead of regularization to deal with overfitting. It has also consisted of data augmentation, SGD with momentum. Oscar et al. [49] used AlexNet to optimize their algorithm.

4.3.2. VGG-16

Simonyan et al. [28] invented the VGG-16, which has three fully connected and 13 convolution layers, carrying the ReLU tradition with AlexNet. VGG-19 is the more in-depth version of VGG-16. Wang et al. used VGG-16 with Graph Convolution Network (GCN).

4.3.3. GoogleNet (Inception-v1)

GoogleNet, also known as Inception-v1 [26]. This architecture consisted of image distortions, RMSprop, and batch normalization. Until adding another layer, the network has a 1×1 Convolutional layer, typically used to minimize dimensionality. Additionally, the global average pooling is used at the network end instead of fully connected layers. Oscar et al. [49] used GoogleNet to obtain feature maps from the annotated datasets.

4.3.4. Inception-v3

Inception-v3 [27] is the third version of Google's Inception CNN. InceptionNet-v3 introduced several new procedures such as RMSProp Optimizer, Factorized 7×7 convolutions, BatchNorm in the Auxiliary Classifiers, and Label Smoothing. Factorizing Convolutions decrease the number of parameters without reducing network efficiency. Label smoothing prevents overfitting. Lucas et al. [52] used Inception-v3 as the base architecture, whereas Egevad et al. [64] used it for transfer learning.

4.3.5. ResNet

ResNet [29] is one of the pioneers of batch normalization. ResNet introduced the first skip connection concept, which allowed the model to learn an identity function. The Identity function ensures the higher layer performs at least as well as the lower layer, not worse. ResNet can be designed even deeper CNNs (up to 152 layers) without compromising the model's generalization power. Kwak et al. [68] used ResNet for feature extracting purposes.

4.3.6. UNet

UNet [30] was first designed and implemented in 2015 to process biomedical images. This architecture consists of three parts: contraction, bottleneck, and expansion. There are several contraction blocks in the contraction section, and each one takes the input using two 3×3 convolution layers followed by a 2×2 max pooling. UNet uses the same feature maps to extend a vector to a segmented image for a contraction. An encoder-like contracting path is used to capture context through a compact feature map. The bottleneck layer interferes between contraction and expansion layers. The expansion layer is like a decoder that makes precise localization. Many researchers used UNet or modifications of UNet for nuclei segmentation tasks [58,65–67,70].

4.3.7. MobileNet

MobileNet [32] is a lightweight but robust architecture to extract features. It has smaller neural networks, low latency, the low computational cost with high accuracy. This architecture has depthwise separable convolutions. Pointwise convolutions follow depthwise separable convolutions. Arvaniti et al. [48] used the MobileNet model to extract features from the images.

4.3.8. Graph convolutional networks (GCNs)

GCNs [33] is a compelling neural network architecture for machine learning on graphs. A randomly initiated 2-layer GCN can produce significant feature representations and leverage its structural information in networks. GCNs draws on the concept of CNN, redefining them for the graph domain. The significant difference between CNNs and GCNs is that CNNs are specially built to operate on regular (Euclidean) structured data, while GCNs is the generalized version of CNN's where the numbers of nodes connections vary, and the nodes are unordered. Wang et al. [73] used graph convolutional networks to propose a weakly-supervised Gleason grading method in tissue microarrays (TMA).

5. Dataset analysis

5.1. PANDA challenge: prostate cANcer graDe assessment using the Gleason grading system [34]

The PANDA challenge is the largest public whole-slide image dataset, approximately eight times the size of the CAMELYON17 [35] challenge. Computational Pathology Group (CPG) of the Radboud University Medical Center and Karolinska Institute's Department of Medical Epidemiology and Biostatistics (MEB) developed this vast dataset. Training set consists of around 11,000 whole-slide images of digitized H&E-stained biopsies. Patients with a pathologist report between 2012 and 2017 were eligible for inclusion. All slides were scanned at 20x magnification and converted to TIFF format. The public and private test set also have a total of 800 whole-slide images. A single case corresponds to a single biopsy (tissue specimen). Multiple cases can correspond to the same patient, but patients from the test set are independent of the training set's patients. The training set contains label noise. This label noise is introduced for several reasons, including inconclusive pathologist reports, annotation errors, errors in the original diagnosis, disagreement between pathologists. For the Radboudumc data and Karolinska data, each of the test sets were graded independently by separate three pathologists subspecialized in uropathology. Due to the subjective nature of the GLEASON grading, some errors can still be present. (See Fig. 3) for a sample image of the panda challenge.

5.2. GLEASON 2019 challenge [36]

GLEASON 2019 challenge dataset is another dataset from the Gleason 2019 challenge, part of the grand challenge for pathology at MICCAI 2019. The dataset is consisting of a set of tissue micro-array (TMA) images. The training set contains 244 Prostate Tissue Microarray (TMA) images, and the test set contains 87 Prostate Tissue Microarray (TMA) images. These TMA images were collected from Vancouver Prostate Center. Each TMA image is annotated in detail by several expert pathologists with level Benign, Gleason grade 3, Gleason grade 4, and Gleason grade 5. (See Fig. 4) for a sample image of the Gleason 2019 challenge.

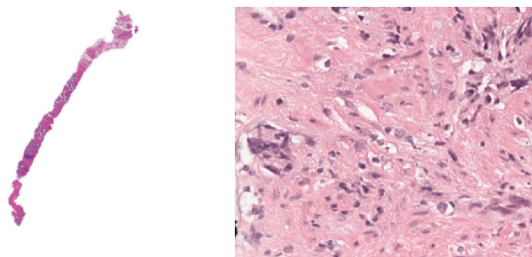


Fig. 3. Sample image of panda challenge.

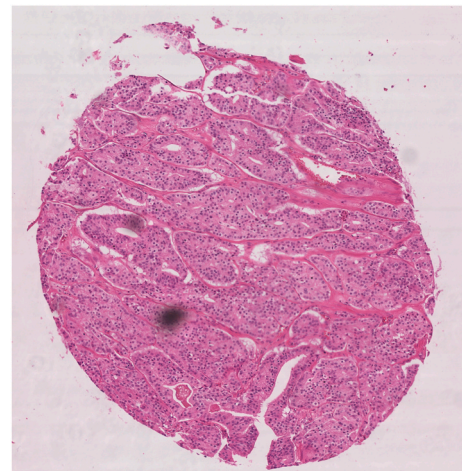


Fig. 4. Sample image of Gleason 2019 challenge.

5.3. TCGA datasets

The Cancer Genome Atlas program is one of the leading and largest databases on earth. TCGA collected and generated a considerable amount of data, including 2.5 petabytes for different types of cancer based on genomic, epigenomic, transcriptomic, and proteomic data. There are different prostate cancer datasets collected at different times.

5.3.1. TCGA-PRAD [38,80]

TCGA Prostate adenocarcinoma dataset contains data from 500 cases. This dataset contains clinical data belonging to this study. There are 490 patient cases. Biospecimen was done on all the 500 cases. Tissues for TCGA were obtained from several locations worldwide to meet their accrual goals, usually around 500 specimens per cancer type. Therefore, the image data sets in terms of scanner modalities, manufacturing, and processing protocols are often highly heterogeneous.

5.4. PESO dataset [86]

The PESO dataset consists of 102 whole-slide images, split into training and test parts collected at different times from Radboud university medical center (Radboudumc) between 2006 and 2011 (IRB number 2016–2275). The training dataset contains 62 WSI, 62 raw color deconvolution masks, 25 color deconvolution masks, and 62 training masks. The test set consists of 40 WSI, 40 XML files containing 160 annotations, 160 png files corresponds to one test region, one padded png files of the same test region, and a mapping CSV file describing whether a test region contains cancer or only benign tissue. (See Table 1)

6. Histopathological image pre-processing and post-processing methods

Histopathological images are quite large and have a complex structure. For this reason, they are challenging for machine learning algorithms. Because these high-resolution images contain much information about image texture, they provide the most successful diagnosis for almost all types of cancer [92]. Histopathological image processing is a method to perform image processing operations to get an enhanced image or extract some helpful information. Different image processing techniques can be used to analyze these images for disease diagnosis and prognosis. Some techniques which are used in histopathological image processing include: pre-processing, post-processing, segmentation, sampling, feature extraction, classification, selection, etc. (Table 2)

Many researchers have performed successful studies using different processing algorithms. According to the study [94], Aureon Biosciences Corporation has developed a proprietary image analysis system

Table 1

Summary of datasets.

Name	Key Features of Image data	Size of Dataset	Accessibility	Dataset Link
PANDA challenge: Prostate cANcer graDe Assessment using the Gleason grading. system [34]	Prostate glands are individually labeled. Valid values are unknown stroma (connective tissue, non-epithelium tissue), stroma and epithelium combined, benign epithelium, the cancerous epithelium (Gleason score 3,4,5).	Training set: $\pm 11,000$ cases Public test set: ± 400 cases (with expert gradings) Private test set: ± 400 cases (with expert gradings)	Open access	http://tiny.cc/w4fosz
TCGA-PRAD	Whole-slide images of prostatectomies collected through microscopic examination of tumor morphology	A total of 771 tissue slides from 480 cases.	Open access	https://wiki.cancerimagingarchive.net/display/Public/TCGA-PRAD
GLEASON 2019 Challenge	Prostate Tissue Microarray (TMA) images were collected from Vancouver Prostate Center. Each TMA image is annotated in detail by several expert pathologists with level Benign, Gleason grade 3, Gleason grade 4, and Gleason grade 5	The training set contains 244 Prostate Tissue Microarray (TMA) images, and the test set contains 87 Prostate Tissue Microarray (TMA) images	Open access	https://gleason2019.grand-challenge.org/Register/
PESO Dataset	The PESO dataset consists of 102 whole-slide images. Of all tissue blocks, 24% contained a region with grade 2, 69% with grade 3, 63% with grade 4, and 33% with grade 5.	The training set contains 62 WSI, 62 Raw color deconvolution masks, 25 color deconvolution masks, 62 training masks. The test set contains 40 WSI 40 XML containing 160 annotations, 160 png, 160 padded png, and a mapping file	Open access	https://zenodo.org/record/1485967#.YEuMn-OzaV4

Table 2

Overview of histopathological image analysis [93].

Processing and Segmentation	Feature Extraction and Selection	Disease Detection, Classification and Post Processing
- Color illumination Normalization - Smoothing - Denosing - Thresholding - Edge-based - Active Contour - Clustering	- Morphometry - Color - Textural - Intensity-based - Morphological - Linear, Non-Linear Feature Reduction	- Supervised - Unsupervised - Neural network - k-nearest neighborhood - Fuzzy system - Morphological Functions

(MAGICTM) using advanced image processing techniques, which is designed to take a single H&E stained image of prostate tissue and extract several raw measurements (spectral, shape, etc.) of histopathological objects as well as spatial associations between them.

6.1. Pre-processing

Analysis of high-resolution histopathological images is very time-consuming. Simultaneously, the complexity of the background and disturbing factors can slow down the processing speed. Image pre-processing algorithms help prevent this unwanted situation. The ability to obtain appropriate features depends on it. For this reason, the proper conditions under which the image processing algorithm can operate as the first job must be specified. Image processing algorithms are adversely affected by noise and various illumination fluctuations. If these negative factors are eliminated, success will increase. In pre-processing histopathological images, techniques like image enhancing, noise reduction, sharpening, tiling, normalization, resolution reduction, stain normalization, ROI detection, morphological operation, etc. used commonly. Pre-processing methods regulate brightness and contrast variations in the image and suppress noise. This provides ease of operation for classification algorithms that are very sensitive to brightness and contrast fluctuations. Several pre-processing methods have been proposed in recent years to reduce the artifacts caused by manual slide preparation. Let us talk about some pre-processing techniques.

Image enhancement is a useful pre-processing technique for feature extraction. For WSIs, different contrast enhancement filters can be applied to remove gray areas by using different OpenCV functions like used add Weighted [95] for this purpose.

Unsharp masking [96,97] also enhances the high-frequency details, and this can be implemented using a gaussian filter and subtracting it

from the original image (See Fig. 5).

Histogram equalization [98] is another image enhancement technique employed to enhance the image. However, in the case of contrasted enhanced images, it did not work [95].

The main problem of working with WSIs is dealing with extremely high-resolution images and large areas of empty space. Moreover, those images consist of large empty areas, leading to GPU memory and GPU time's inefficient use. So, effective locating the areas of concern and zooming them in would be the key to reach better performance. This problem can be easily solved using the Tiling method [99] and Concatenate Tile pooling method [100] (See Fig. 6).

Color normalization has become a necessary step in pre-processing because the staining procedure differs markedly between training and testing images. Researchers use color normalization to reverse the impact of unintended improvements in tissue images due to staining and scanning variations across hospitals. However, according to the study [65], it is better to use a source image than a color-normalized image (See Fig. 7).

As we said earlier, color normalization of stained tissue samples is one of the most critical pre-processing algorithms in the whole-slide picture (WSI) analysis [101]. Despite the standardized staining protocols, variations in staining outcomes may occur due to differences in antigen concentration, incubation time and temperature, slide scanner conditions, and so on [102]. The efficiency and accuracy of CAD structures can be affected by such color/intensity differences. Stain normalization methods aim to aid CAD schemes by producing photographs of a consistent appearance of the various strains [103,104]. Researchers use different types of methods to build a high-performance stain normalization system. In this paper [105], they used the Macenko

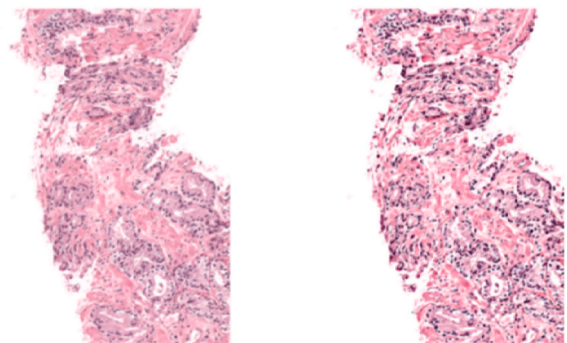


Fig. 5. Before(left) and after the (right) result of unsharp masking [97].

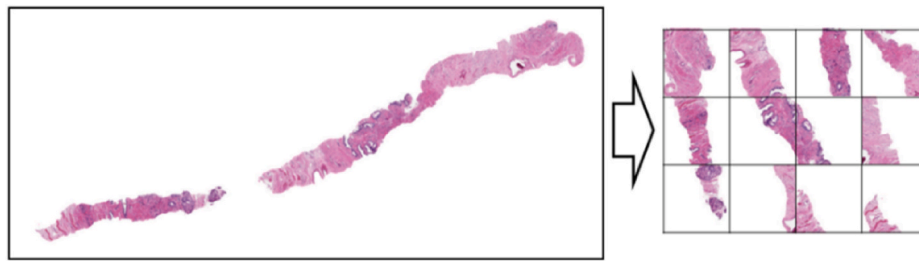


Fig. 6. Tiling method [99].

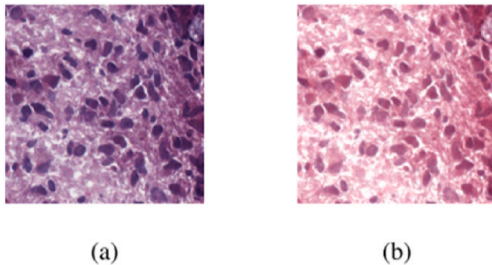


Fig. 7. Color normalization visualization, (a) original image, (b) normalized image. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

method [106].

Another pre-processing strategy used to minimize processing time is the region of interest (ROI) detection [108]. In some frameworks, ROI detection and noise reduction are often performed simultaneously. For example, the pre-processing stage for tissue-level feature computation selects the ROI by removing regions with little content and noise [109]. Noise reduction is accompanied by ROI detection to assess the nuclei area for nuclei level function computation [110].

In most cases, deep learning algorithms are used on whole biopsies or WSIs. However, there are certain limitations to integrating CNN into a WSI. To begin with, discriminative data may be lost due to the need for detailed image downsampling. Second, CNN could potentially learn from only one of the many discriminative patterns in the image, which would result in data inefficiency [107]. The discriminative information in histological images is encoded in high-resolution patches; thus, the trick is to train the network on high-resolution patches and then predict the entire WSI label based on patch-level predictions. The models are taught directly from the data using deep learning techniques. It is of utmost importance to select representative patches of the image for training [107] (See Figs. 8 and 9).

Nucleus segmentation is another important pre-processing task since many of the characteristics of diseases, especially cancer, are expressed in the cell nuclei [111]. Indeed, most cytological and histopathological analysis is based solely on nuclear features. On the other hand, Nuclei segmentation is a complex task: diverse tissue types, staining variations, and cell types all express different visual characteristics, making it challenging to develop conventional image segmentation algorithms that perform well in all these situations [112].

There are numerous methods through which histopathological images can be pre-processed. According to Ref. [81], we can say that every pre-processing method has its effects on CNN. Using the same

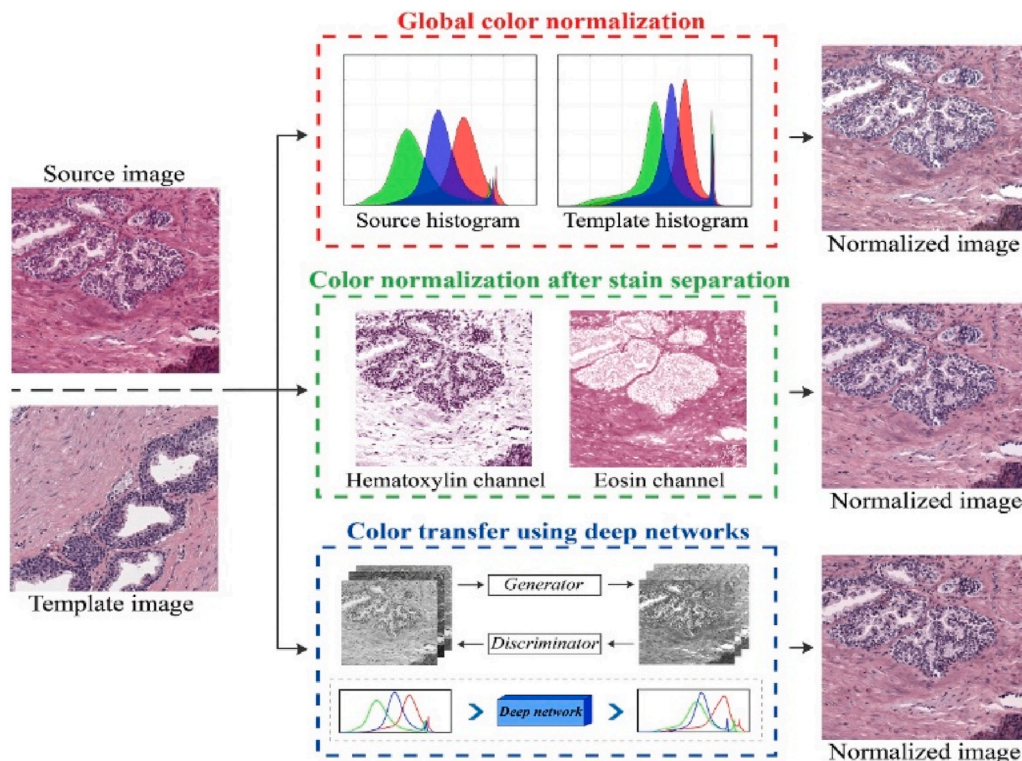


Fig. 8. Stain normalization strategies in deep learning frameworks [107].

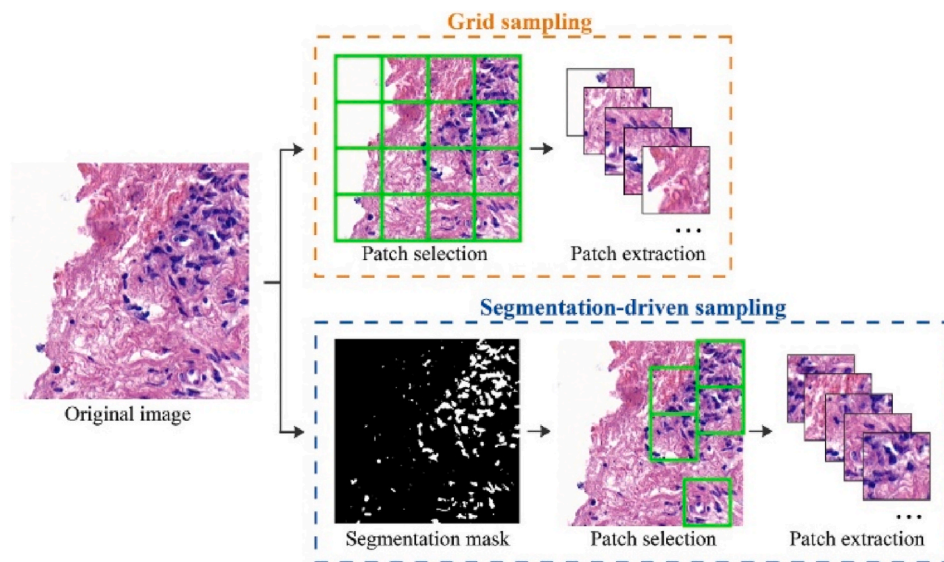


Fig. 9. Patch selection techniques in deep learning frameworks [107].

histopathological image for three different pre-processing methods, the study [81] has shown the effect of different pre-processing methods on Convolutional Neural Networks.

6.2. Post-processing

Researchers use different post-processing techniques for further improvement of the histopathological image Analysis. In post-processing histopathological images, techniques like active contour differentiation, color illumination, image compression, image restoration, zooming, augmentation, etc., are used commonly. Generally, the post-processing approach is used for classification, detection, and segmentation tasks. As we can see in Ref. [107], The accuracy and the performance improved dramatically after using different post-processing methods. For example, after using CNN + majority voting, the accuracy improved by 89.2% in Gleason grading classification [113]. By using ResNet with scale embedding, the performance improved by 95.3% accuracy in cancer detection. 1.9% accuracy improvement compared to the single network [114] (See Fig. 10).

Semantic segmentation is a well-known computer vision issue, and it is especially relevant in medical imaging research [65]. To avoid this problem, the algorithm mentioned in Ref. [65] needs some complex

post-processing methods. Different post-processing techniques like Morphological operations are also used to remove artifacts [67]. According to the study [67], we can see the effects of post-processing methods in Table 3.

From the future perspective, post-processing methods will be very impactful. According to a study [52], although the patch-based classification can be considered accurate, classification results may be improved by introducing extensive post-processing.

6.3. Staining process of histopathology image

Histopathology is the microscopic examination of a biopsy or surgical specimen that has been obtained and fixed on glass slides to

Table 3

Segmentation Performance Comparison for Different Methods used in [67].

Segmentation Performance Comparison for Different Methods			
	Precision	Recall	F1 Score
Segment Using CNN without post-processing	0.8823	0.8235	0.8453
Segment Using CNN with post-processing	0.8921	0.8123	0.8460

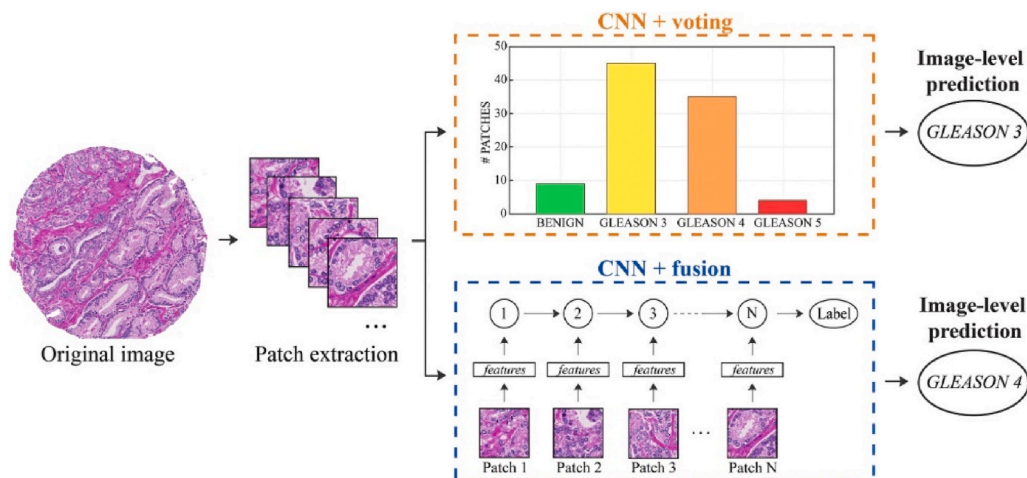


Fig. 10. CNN + voting: for each classified patch, a 'voting' procedure is applied to determine the entire image's final class. CNN + fusion: integrates a decision model to aggregate each patch's classification and obtain the label of the entire image [107].

examine disease symptoms. The pieces are dyed with one or more stains to visualize various tissue elements under a microscope. Staining is used to highlight essential features of the tissue as well as to enhance the tissue contrast. Staining aims to expose cellular components; counter-stains provide the contrast as it allows for identifying defects in cell count and structure under the microscope. From dyes and metals to labeled antibodies, histology uses a wide range of stains. Metachromasia is a disorder in which certain stains cause cells and tissues to change color significantly from the pigment complex's original color. Typically, paraffin-embedded tissue fragments are used for staining. They are rehydrated and stained after being cleared (transparent) with a clearing agent like xylene. However, there are several staining techniques used in histopathological image processing.

Hematoxylin and Eosin (H&E) staining Fig. 11 is the most used staining process in histopathology. Pathologists have used Hematoxylin-Eosin (H&E) staining for over a hundred years. It provides the pathologist/researcher a very detailed view of the tissue. It achieves this by clearly staining cell structures, including the cytoplasm, nucleus, and organelles, and extra-cellular components. Two dyes, Hematoxylin and Eosin, are used in the H&E stain. Hematoxylin reacts like a basic dye with a purplish-blue. It stains acidic, and eosin is an acidic dye that is typically reddish or pink. It stains basic [84].

PAS staining: Periodic acid-Schiff is a special histology stain used to mark carbohydrates (glycogen, glycoprotein, proteoglycans) Fig. 12. PAS is commonly used on liver tissue where glycogen deposits are made, which is done in efforts to distinguish different types of glycogen storage diseases [84].

Acid Fast Staining: The acid-fast stain is a differential stain that is used to distinguish acid-fast species like *Mycobacterium* representatives. Wax-like, nearly impermeable cell walls characterize acid-fast organisms; They produce significant quantities of fatty acids, waxes, and complex lipids, as well as mycolic acid. Acid-Fast Stain is an example of *Mycobacterium tuberculosis* (See Fig. 13).

Van Gieson: The van Gieson stain is a common stain for separating collagen from other connective tissue, such as muscle tissue. It is a term that's often used to describe the fiber structure in different types of tumors. The stain penetrates the tissue sample in a mixture of picric acid and acid fuchsin, causing collagen to turn red. The muscle tissues and blood cells in the area have been painted yellow (See Fig. 14).

Gomori Trichrome: Gomori Trichrome (blue) (submucosa). Trichrome stains are used to stain and identify muscle fibers, collagen, and nuclei. They can be used to contrast skeletal, cardiac, or smooth muscle [116] (See Fig. 15).

7. Problems specific to prostate cancer image analysis

Research on Deep learning-based prostate cancer detection and Gleason grading is rising day after day. Researchers are working hard to detect prostate cancer and improve the Gleason grading method all over the world. Nevertheless, a variety of drawbacks remain. Researchers have to deal with these limitations and improve the system.

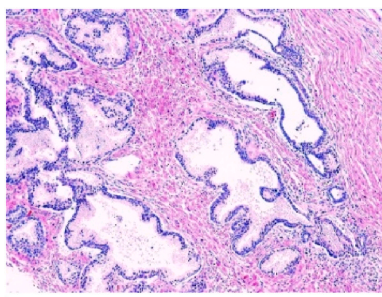


Fig. 11. Prostate H&E stain [84].

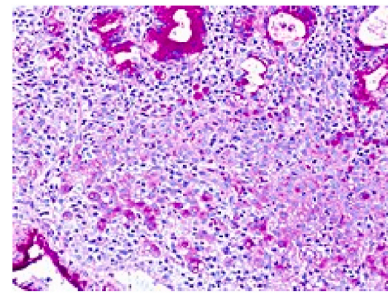


Fig. 12. Periodic acid-Schiff (PAS) stain [84].

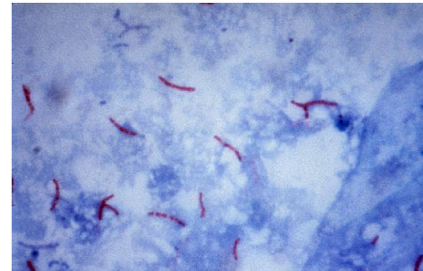


Fig. 13. Mycobacterium tuberculosis visualization using the Ziehl-Neelsen stain [84].



Fig. 14. Van Gieson stain [117].

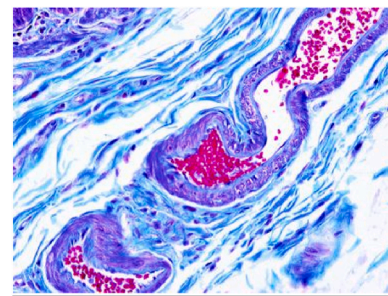


Fig. 15. Gomori trichrome (blue) (submucosa) [116].

• Limited dataset and privacy issue

As a consequence of the privacy issue, there are limited data sets available on prostate cancer. In order to ensure data authenticity, several experts must conduct all images annotation. Data shortage complicates training and can lead to overfitting.

- **Gigapixel image size**

Prostate tissue microscopic whole slide images are in gigapixel. When our input images' size is too large, more parameters are needed to estimate, which leads to more computational power and memory. So, it is hard to train and detect such a large size image. By resizing the entire image into several image patches, we can get a better solution. However, determine a perfect patch size is a very complex task.

- **Various images/sequence demonstrate multiple elements of prostate cancer**

Various whole slide images display various facets of prostate cancer and require an appropriate way to combine independent information. A T2-weighted sequence is suitable for the description of the zonal prostate anatomy. It may be used to examine the prostate fossa and seminal vesicles in-depth [41]. The apparent diffusion coefficient (ADC) and prostate carcinoma Gleason score negatively correlates [40].

- **Limitations of Transfer learning**

The transfer learning approach can be used to resolve the lack of data set issues. The lack of initial and target problem similarity, transfer learning can lead to negative transfer. The conventional transfer learning model considers each image separately without exchanging details on the intra-category correlation. It is hard to remove layers with confidence to reduce the number of parameters because of its nature, which finds low-level features. Densely connected layers and deep convolutional layers can be good points for a reduction, but it difficult to see how many layers and neurons to reduce to avoid overfitting [43].

- **Inter-observer variability**

Inter-observer variation is the difference between the results attained by two or more observers studying the same thing. Intra-observer variation happens in PCa diagnosis when two or more pathologists look at the biopsy and put different options. One of the researchers' problems is that proper and accurate manual grading was not always possible [64]. Misinterpretation can impact the reproducibility of the system. Computer-aided systems or Artificial intelligence help us to identify the Gleason scores faster and let us take the decision easily.

- **Different Levels of Magnification Result in Different Levels of Information**

Between two types of information, cell structure and glandular structure, the cell structure is clearly visualized in the high-power field (HPF) microscopic images, and glandular structure is clearly visualized in the low-power field (LPF) microscopic images. Cancerous tissue concludes both cellular and structural atypia; therefore, images at multiple magnifications are essential for research work. Sometimes inputting both low and high magnification images simultaneously to model gives better accuracy.

- **Color variation and Artifacts**

While slicing and placing the pathology specimens on a slide-glass with hematoxylin and eosin, some undesirable effects could be introduced, like tissue deformation and wrinkled, and dust can be mixed with slides. As these artifacts can change the actual output, some algorithms like blur [90] and tissue-folds [91] have been introduced.

According to Daisuke Komura et al. [88], color variation is another serious artifact. This variation occurs due to different manufacturers of staining reagents, staining conditions, the thickness of tissue section, scanner models, etc. Consideration of color variation can help us getting better accuracy. For this, we need enough data on every stained tissue

from every scanner.

- **Positive bias and ROI-Align layer**

Wenyuan et al. [44] noticed that 5-fold validation is not a patient-wise validation as they did not have patient-level information. This might result in a positive bias as cancer can notice similarities in tiles within the same patient, especially in closely related tiles.

According to this paper, The ROI-Align layer extracts a small feature map from the corresponding feature pyramid layer for each ROI right before the network head. It loses some crucial histopathological information. This type of information is critical for Gleason grading as the sizes of glands are different.

- **Existence of different types of tumor and foreign tissue**

In Wouter Bulten et al. [89], they noticed that other tumor types and tissue could be present in prostate biopsies, e.g., colon glands should be excluded for Gleason grading. Other prognostic information can be presented in the biopsies. As an example, detection of intraductal carcinoma is prognostically relevant.

- **Excessive Pre-processing**

In Öztürk et al. [81], When the results are examined, it has been seen that pre-processing methods contribute to learning to a certain extent. This contribution varies depending on the cleaning of the noises and the drying of the properties. If pre-processing is excessive, success cannot reach the desired level. For this, it is crucial to pre-process images appropriately.

8. Model

8.1. Region of interest-based models

Accurate localization of small areas of the region of interest is a challenging task in histopathological images. Wenyuan et al. [44] proposed an ROI-based model where they extracted feature maps for each class using ResNet from the dataset images. Then the feature maps are fed into two heads: 1) Epithelial Proposal Head (EPH) and 2) Grading Network Head (GNH). EPH identifies whether the image includes any epithelial tissue or not. Region Proposal Network (RPN) recognizes regions for the Grading Network Head (GNH). GNH assigns Gleason grades to these regions. If no epithelial cells remain, the model considers the whole image as the stroma. Whereas Oscar et al. [49], used different state-of-the-art architecture like LeNet [51], AlexNet [25], and GoogleNet [50] to train the annotated datasets. Each architecture was trained using the same patches obtained from the previous patch selection step with 40x resolution. Various image augmentation techniques on patches were used, focusing on the region of interest. In higher Gleason grading cases, they used heatmap, the region of interest (ROI) in the images.

8.2. Segmentation based models

In the paper of Wenyuan et al., Grading Network Head (GNH) is a nuclei process of segmentation task [44]. Most authors have used the different versions of UNet for the segmentation of nuclei in histopathological images. Zeng et al. [65] Zeng et al. [65] have used a series of Residual-inception (RI) blocks to isolate multiple local features during the encoding process. They used a model similar to UNet to concatenate high-level features with low-level features in the decoding phase. Its configuration contains three components and two outputs. The left component is similar to the UNet architecture up-sampling portion, and the output is an abounding mask for the nuclei. Bulten et al. [58] proposed an extended version of UNet. They used CycleGan for

normalizations of images and variations. Oskal et al. [66] used UNet as their underlying architecture for epidermal tissue segmentation. The design involves a contracting direction to four stages. Every layer has two operations of 3×3 convolution, with the activation of a rectified linear unit (ReLU). A max-pooling sheet accompanies the rectified linear unit. They treated localization cases in the network's four-stage decoder module. They treated cases of localization of nuclei at this point. The feature map here is upsampled and concatenated with the corresponding better resolution feature map. It blends semantic details with specific localization of region of interest. They applied additional convolutional operations to the concatenated feature maps at each point, which create a more reliable production. Usage of a final 1×1 convolution to move the last function chart to the ideal one. They modified UNet and used zero-padded convolutions on the convolution row, except the last 1×1 convolution. The ReLU activations were also moved to the Exponential Linear Unit (ELU) [60]. ELU is a saturating hyperparameter with negative values. After that, for any convolution row, they applied batch normalization [63]. Batch normalization provides the model with an impact on the regularization. In Ren et al. [67], their CNN architecture consists of an encoding and decoding network. Both encoding and decoding layers have ten convolutional layers. The convolutional layers have a kernel size of 3×3 . A Rectified Linear Unit follows each convolutional layer, batch normalization layer, and a 2×2 max pooling layer with stride 2. In the decoding module, they used the upsampling method instead of max-pooling. The upsampling layer uses the max-pooling layer's location to reverse the max pooling operation with stride 2. The final layer has a softmax classifier. The softmax classifier was used for the binary classification. Oda et al. [70] proposed a fully convolutional neural network for segmentation. A semantic segmentation approach was used to get a better result. They aimed to train the contour specified boundaries and use affirmations to assess maximum adaptive segmentation training loss. Batch normalization and ReLU activation functions were used in the model after two layers of 3×3 convolutions, with two max-pooling per stage in the encoder network. Their model was as follows: convolution, batch normalization, ReLU, convolution, batch normalization, ReLU, and MP 3 times. There were two decoding paths in this sequence. It is an improved version of UNet. In BESNet there are 1) Boundary Decoding Path (BDP) 2) To Decode feature maps. The feature maps which were obtained from this method were used for segmentation purposes.

8.3. Mobile architectures

Arvaniti et al. [48] used MobileNet as their base architecture. They used the MobileNet model to extract significant spatial features from the patches. It helped detect patterns at the regional level and produced probability maps at the pixel level for each Gleason grade. Patches were predicted only if the probability is more significant than 0.8. In this paper, class activation mapping techniques are used with a fully convolutional architecture. MobileNet model can be converted into a Tensorflow lite model. Tensorflow lite models can be easily deployed on Android or Apple devices.

8.4. Fusion of CNN and KNN

In Nagpal et al. [47], the model has two stages. At the first stage, they extracted the features using the ensembled Convolutional Neural Network. In the second stage, they used the Kth-nearest-neighbor algorithm to group different Gleason grades. In the first stage, the model was trained using region-level features. Applying an independent, partially trained model to the entire training dataset, they applied hard negative mining. In the second stage, they collected the labels of each image from the pathologists. They applied Kth-nearest-neighbor (KNN) algorithm to predict the Gleason Grades (1, 2, 3, or 4-5). KNN has been used to Gleason grades concerning Pathologists' manual Gleason scoring [73,78]. Hatipoglu et al. [71] have used SVM and KNN for such purposes

too.

8.5. Transfer learning approach using state-of-the-art architecture

Lucas et al. [52] retrained Inception-v3 [53] CNN design was used to distinguish Gleason Grade 3 and Gleason Grade higher than or equivalent to 4 from non-typical various prostate histopathology tissue. The design was re-engineered using CNTK, the toolkit used for image recognition in deep learning [54]. In contrast, Kallen et al. [55] used a novel deep Convolutional network pre-trained on a large image dataset. They designed an algorithm to mimic how pathologists Gleason scores manually. There are 22 layers in their network architecture. The final output from the final layer was used to train a random forest classifier and a Support Vector Machine (SVM) classifier. In Asami et al. [57], they proposed a novel Deep Convolutional Neural Network-based model to extract features from the histopathological images. Their model consists of 3 convolution layers. Each followed by a pooling layer and a TanH. They used 5×5 filters for convolution layers and 2×2 for pooling layers. A classifier was used to group images into their respective grading. Their model takes 100×100 pixels as input and extracts a probability map at the final layer. They used MxNet [78], a deep learning library, to build their model. In Egevad et al. [64], they built an artificial intelligence system for comparing and solving the Gleason grading complexities. Their architecture consisted of an ensemble of two convolutional deep neural networks. Each network was composed of 30 Inceptionv3 models pre-trained on ImageNet. The last layer, otherwise known as the classification layers, tailored according to its size. The rectified linear unit is used as the role of neuron activation. The pixel in the resulting image contains the sum of filter kernel products and the input image's corresponding local area. The input image from the other layer is entangled with the 2-pixel stride filter kernels.

8.6. Novel CNN approach

Anna et al. [72] proposed a novel architecture of the CNN to gain spatial details from the images. The network output consists of four channels, one for each of the four classes. They are normal-tissue, Gleason grade 3, 4 and 5. On the last two layers, they used dropout as regularization. They decreased the learning rate continuously to achieve a consistent local minimum. They implemented their architecture on MatConvNet11 [31]. Kwak et al. [68] built a novel architecture using CNN nuclear seed maps. The architecture has six layers of ANN (Artificial Neural Networks). Primarily three convolution layers, two fully connected layers, and a SoftMax layer. In Neeraj et al. [69], they proposed a CNN Architecture for predicting the distance transform of nuclei from the images. In Hatipoglu et al. [71], they proposed a novel pyramidal convolutional neural network. Their model consists of 4 convolutional layers, followed by four subsampling layers. The third level has six convolutional layers, and the fourth level has six sub-sampling layers, respectively. In the final layer, they used a fully connected dense neural network for classification. Then, before adding to a bias term, they multiplied the sum by an adjustable weight.

8.7. Unique graph convolutional approach

Wang et al. [73] made an exceptional contribution by proposing a novel method using the Graph Convolution network.

They made a graph representation of nuclei or cells as nodes. Each node can accumulate feature vectors from its neighboring nodes through iteration. In GCN, graphs can generate a new feature vector at the next hidden layer of the network. Morphological features, texture features, and Contrastive predictive coding are extracted for each nucleus. Accordingly, for each feature, GCNs can learn to represent in a node. It works like the feed-forward method as on CNN. They obtained a unique graph representation after max pooling.

Table 4 represents a comparison of different deep learning-based

Table 4
Summary of the significant papers.

Reference	Application	Deep learning architecture	Training	Patch size (pixels)	Accuracy	Dataset
Wenyuan et al. [44]	Two-step training methods for the identification of epithelial cells and Gleason grading simultaneously	R-CNN	end-to-end	512×512	Epithelial cells detection accuracy 99.07% Gleason grading (overall pixel) accuracy 89.40%	Privately owned
Nagpal et al. [47]	Gleason scoring of whole slide images of prostatectomies	CNN	end-to-end	911×911	70% accuracy on Gleason scoring task	TCGA Dataset
Arvaniti et al. [48]	Heterogeneous Gleason grading	MobileNet	Transfer learning	750×750	65% accuracy on evaluation	Privately owned
Oscar et al. [49]	Using transfer learning methods to classify whole slide images considering the region of interest	GoogleNet	Transfer learning	128×128 , 256×256	40% of the patches classified as high grading among 36 out of 46 images with a 78.2% accuracy	TCGA-PRAD Dataset
Lucas et al. [52]	Differentiating GP 3 and GP ≥ 4 from non-atypical tissue	Inception-v3	end-to-end	299×299	- Non-atypical and malignant (GP ≥ 3) accuracy 92% - Differentiation between GP ≥ 4 and GP ≤ 3 accuracy 90% (sensitivity 77% and specificity 94%)	Privately owned
Kallen et al. [55]	Classification of both patches and whole slide images	Deep CNN	end-to-end	87×87 , 167×167	The architecture was designed for Gleason, scoring from 3 to 5. The model provides an accuracy of 81.1% among the patches. Classification of entire images provides an accuracy of 89.2%.	TCGA Dataset
Asami et al. [57]	Disease stage classification method for Glioma histopathological images	Deep CNN	end-to-end	28×28	Training and classification accuracies were 98.2% and 87.2%	TCGA Dataset
Bulten et al. [58]	Automated Gleason grading from prostate biopsies comparing with the pathologists' annotations	UNet	end-to-end	–	AUC for benign vs malignant is 0.990, 95% CI (0.982–0.996)	Radbound University Medical Center Dataset
Egevad et al. [64]	Identification of areas of Gleason scoring difficulties and comparison of AI system with pathologists' image	InceptionV3	Transfer learning	2048×2048 (Whole image)	Gleason score of $3 + 3 = 6$, $3 + 4 = 7$, $4 + 3 = 7$, $4 + 4 = 8$, and $9 - 10$ in 13.9% (5), 25.0% (9), 33.3% (12), 19.4% (7), and 8.3% (3) respectively	Pathology Imagebase (ISUP) Dataset
Zeng et al. [65]	Nuclei segmentation using residual inception channel attention UNet architecture	RIC-UNet	end-to-end	224×224	The average DICE [74] coefficient is 0.8624	TCGA Dataset
Oskal et al. [66]	Epidermal tissue segmentation	UNet	end-to-end	512×512	- Training dataset mean Positive Predictive Value 0.84 ± 0.28 (Sensitivity 0.97 ± 0.07) - Test dataset mean Positive Predictive Value 0.89 ± 0.16	University of British Columbia Virtual Slidebox [61]
Ren et al. [67]	Computer-aided Imaging method for Gleason pattern 3 and 4 classification using nuclei segmentation technique	UNet	end-to-end	480×360	The sensitivity, specificity, and accuracy were 0.70 ± 0.15 , 0.89 ± 0.04 and 0.83 ± 0.03	Privately owned
Kwak et al. [68]	Localization of epithelial nuclei in prostate tissue microarrays	CNN	end-to-end	128×128	AUC of 0.974 with 95% Confidence Interval	Tissue microarray research program at the National Institutes of Health dataset
Neeraj et al. [69]	Prostate cancer recurrence prediction using prostate tissue images	CNN	end-to-end	51×51	0.81 under AUC curve	CPCTR Dataset
Oda et al. [70]	Detection and semantic segmentation of cells using enhanced boundary detection architecture images	UNet	end-to-end	1636×1088 (Whole image)	89.5% of the detection rate	Nagoya University Hospital Dataset
Hatipoglu et al. [71]	Classification of biopsy images with spatial dependencies of cell and non-cell pixels evaluation	CNN, SVM, KNN	end-to-end	896×768 (Whole image)	The best accuracy was 86.91%	Yale Tissue Microarray Facility Dataset
Anna et al. [72]	Gleason grading using small convolutional network into grade 3, 4, and 5	CNN	end-to-end	106×106	92.7% accuracy	Dataset from PathXL, Belfast, and Beaumont Hospital in Dublin
Wang et al. [73]	The spatial organization of cells as a graph to classify into Gleason grades	GCN	end-to-end	256×256	88.61% accuracy and AUC of 0.9659	BACH dataset [62]

prostate cancer detection & Gleason grading architectures and their used techniques.

9. Discussion and future directions

9.1. Discussion

Automated diagnosis can reduce workloads and pathologist variability [83,84]. Researchers face difficulty in studying the Gleason scoring system. Accurate annotations and pathological accuracy are

required to train the model correctly.

Deep learning models are proven to be robust to a small number of errors in annotations. It is necessary to make prostate cancer models robust and more accurate. For this purpose, a large number of datasets are needed for training and testing. Data collection should be done from different parts of the world to maintain variation in images. Several pathologists should annotate datasets to decrease inter-observer variability [52,58,64,66,67]. The benefits of collecting data from various parts of the world will make the applied methodology more resilient against variations in biopsy appearance and improve CNN performance.

By taking into account more neighborhood information, CRFs can improve data annotation [85]. More open datasets and collaboration, like the Panda challenge [34], will boost the research sector and reduce data unavailability.

Compared to radiological image processing and analysis, histology image processing and analysis is a newer area with more difficult challenges. The structure of histology images is much more complex than that of radiology images, and most importantly, the central challenging part of dealing with histopathological images is dealing with images of extremely high resolution and large areas of space. The findings of the research on various aspects of histopathological image analysis are very promising. However, comparing different histology image methodologies is difficult because each study uses a different image dataset and uses different metrics to display results. Furthermore, the tissue architecture and cell types of different organs vary, so a procedure that works for one organ may not work for another. In terms of pre-processing methods, image enhancing, stain normalization, Tiling is the most widely used digital pathology algorithm. The stain normalization procedure compares the appearance of a source image's stain color to that of a reference image. Due to the standardization of histological images, the deep network should learn not only a particular color distribution but also histopathological patterns. Numerous studies have also shown how including these pre-processing steps gives higher performances and better accuracy [65,101–107]. The image-enhancing technique is very powerful in terms of feature extraction and segmentation. It includes Unsharp masking, Histogram equalization. These techniques enhance the high-frequency details of a WSI. Many studies have also shown how including these pre-processing steps gives better accuracy [95–98].

Patch selection is another critical pre-processing step in CNN training. Different classic algorithms based on thresholding, color deconvolution, and active contour models were used to define the regions of interest to extract the CNN patches. Finally, when processing whole-slide images (WSIs), it is critical to accurately identify histological tissue and the different artifacts during the histological slide preparation process. Several methods for histological tissue segmentation and artifact detection have been suggested over the years, the majority of which rely on color space modification and adaptive thresholding. A CAD program should process a whole slide quicker by deleting background regions and avoiding areas with altered morphology or intensity. Post-Processing is another powerful image processing technique in order to improve accuracy and performance. The most common algorithm adopted in digital pathology is illumination, image compression, image restoration. Simple voting methods (such as max voting) and more sophisticated models (such as random forest or nearest-neighbor classifiers) will be used in these strategies. Many studies have shown how pre-processing steps improve performance [67,107,113,114].

Histological staining is a collection of techniques for preparing sample tissues for examination under the microscope by staining them with histological stains. In fact, staining is used to highlight essential tissue features and improve tissue contrast and reveal cellular components. For structural and architectural analysis, pathologists use staining to isolate cellular components in tissue. In histology, a wide variety of stains are used. Hematoxylin and eosin is a basic dye that is commonly used in this process and stains the nuclei. However, there are several other staining techniques used in histopathological image processing. For example, PAS staining, Acid Fast Staining, Gomori, Trichrome, Van Gieson, Toluidine Blue, Alcian Blue, Giemsa, Reticulin, Nissl, Orcein etc [84,116,117]. Hematoxylin and Eosin (H&E) staining is the most used method. The (H&E) stain is used to distinguish between the nuclei, cytoplasm, and connective tissue of cells. Cell nuclei are stained blue with hematoxylin, and cytoplasm and connective tissue are stained pink with eosin. Other stains include DAB, immune-histochemistry stain, and others.

Still, there are many significant limitations while we deal with prostate cancer detection. We have shown several existing limitations in

this domain in the limitation section. These constraints can be a significant area for any researchers' potential study. Due to privacy policy, hospitals do not publish their patient image dataset. Since prostate cancer is a sensitive issue, patients feel uneasy about sharing their information regarding prostate cancer. It produces difficulties and increases the use of reproducible data. Recently, Kaggle released a vast dataset [34], one of the noteworthy contributions in prostate cancer studies. We have portrayed different deep learning models and how they can detect prostate cancer from complex histopathological images. Image sizes are in gigapixel and hold much information. Sometimes it is very tough to apply transfer learning. Interobserver variability and the existence of different types of tumors also make this process very difficult. Color variation and artifacts sometimes hamper the pre-processing step. The result of the model was affected due to excessive pre-processing. In most of the surveyed papers, researchers use very little data to train their deep learning system. Mostly, researchers face difficulty in detecting the boundaries and contours of cells [65,66]. Tissue deformation and incorrect lab results of the H&E study become a barrier to get better-annotated data [48,58,66,68].

Among all the surveyed papers, Wenyan et al. [44] used a unique deep learning architecture where they first detected if cells are epithelial cells or not. Their model is based on a region proposal and works on the whole image. In fact, they provide a solution to the creation of unnecessary patches from a whole slide image. In some of the papers, researchers engaged a unit of pathologists to verify the model's performance. In fact, traditional machine learning techniques are used to analyze the model's robustness compared with pathologists' decisions [48,52,58,67–69]. Transfer learning is frequently used in detecting both high-level and low-level features fast and efficiently. Fine-tuning is used in different papers to increase accuracy and prevent overfitting [48,64,68]. Among the surveyed papers, pre-trained models on ImageNet are used frequently [55,64,68,72,73]. In most papers, image segmentation and patch processing are necessary to handle super-pixel histopathological images for a better training approach [48,49,52,58,64,65,67,68,68,69,71,73]. Many researchers have used modified UNet for nuclei segmentation. Nuclei segmentation helps to retrieve morphological information from the images [58,65–67,70]. Among them, Zeng et al. [65] have used Residual-Inception UNet. Graph convolution Network is a noteworthy and uncommon approach in deep learning. Wang et al. [73] have used the Graph Convolution network in Gleason grading prostate tissues by organizing nearby cells' spatial information.

9.2. Future directions

Deep learning has demonstrated substantial progress in various implementations relative to many other forms of machine learning. One significant benefit of deep learning is that it eliminates the need for feature engineering. Pre-processing and post-processing of data is a significant and challenging task before applying deep learning techniques. Furthermore, the adaptation or modification of existing deep learning architectures on new applications is relatively easy. Scholars need to keep in mind how their system gives an efficient and better result. If the specificity and sensitivity are not well aligned, it does not make sure a trusted system. In fact, it is essential to determine whether the tissues are malign or benign for the classification of biopsy reports. Later, malign tissues should be graded into various Gleason patterns. There are always some pitfalls of a system. From our surveyed paper, we have understood the following things:

- Deep learning models often require a large amount of training data to achieve superior performance compared to other methods.
- Computation cost is exceptionally high and time-consuming.
- Pathologists' cohort is necessary for proper validation in determining correct Gleason grading for prostate cancer tissues.
- Since whole slide images, i.e., histopathological images, look quite similar, it is not easy to differentiate one from another for the model.

The findings evaluated during this study have utilized various datasets and imaging techniques. Hence a distinction between the efficiency of several evaluation metrics and clinical practices was more complicated. As we see in most of the studies, the pathologists' cohort is used, making it fair to depend on the evaluation results. It shows the ability of deep learning systems in the identification and treatment of prostate cancer.

10. Conclusion

Detection of prostate cancer using whole slide images is a landmark in the area of medical pathology. We analyzed numerous articles on deep learning usage throughout this paper to identify prostate cancer from histopathological images. Machine learning and deep learning have opened the door to medical image studies. There are still many undiscovered fields that need to be explored. In this paper, we discussed basic concepts regarding prostate cancer. Convolutional architectures have increased in usage rate during the past decade for processing complex images. We provided pieces of information about several state-of-the-art techniques surveying from most recent works. We have also discussed Gleason grading methods, histopathological image pre-processing, and post-processing techniques. Data insufficient, super-pixel image, the difference between inter-observer and intra-observer variability have caused researchers to suffer working with histopathological images. In most of the research papers we studied, we have seen limitations and a lack of reproducible data in this research field. We have highlighted the evaluation criteria and metrics for loss calculation in different types of model architecture. This paper shows the pathway of how to use deep learning for prostate cancer detection from histopathological images.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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